Coherence Without Content

A value-coherence metric scores invented outcomes at 0.880, and 6 of 9 models still clear their own floor

Nullcard

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Abstract

Replace the referents of a preference battery with invented words and a language model's choices stay almost as coherent as its choices over real outcomes: 0.880 against 0.906 on the Thurstonian held-out accuracy that has been read as evidence for emergent value systems. The models are not blind to the substitution — a channel of the same forward pass that the metric *discards* separates real outcomes from invented ones at AUROC 0.821, against 0.596 for the channel it keeps. The coherence number is computed from the one channel that noticed least.

Utility Engineering (Mazeika et al., 2025) fits that model to a language model's pairwise choices, reports the accuracy as *structural coherence*, and concludes from its rise with scale that coherent value systems emerge in large language models. Its robustness checks vary how the question is asked — language, syntax, framing, option labels, context — but none varies whether the outcomes being ranked *mean anything.* We build that control: the same 510 outcomes with their referents replaced by invented words, holding the prompt, the pair set, the fit and the metric fixed.

The control bites hardest on the scaling claim. Across one model family from Qwen3.5-0.8B to Qwen3.5-9B, real-outcome coherence rises +0.014 while coherence on *meaningless* outcomes rises +0.085 — roughly six times as fast — so the content-attributable residual *falls* by -0.071 as models grow. What increases with scale is chiefly the component that does not depend on content. The mechanism is visible in the aggregate: the pair above is a mean over 9 models, 5 evaluation splits and three independent design replicates each (81 cells), leaving a residual of +0.025, and it is small because the metric thresholds preferences to hard labels and so records direction while discarding strength — the same models commit to a side on 41% of pairs about real outcomes and 4.5% about invented ones, a median collapse of 17×.

We are explicit that a real semantic signal exists: 6 of 9 models exceed their own design-replicate noise floor. The claim is that this signal is small, that it shrinks as models grow, and that the reported number does not separate it from format. Nor is the instrument merely insensitive: an installed persona displaces real outcomes substantially further than invented ones in 18 of 20 conditions, so content-linked preference structure is detectable when present — and inducible. The blindness does recur one level up: told to have a trait, to hide it, or to perform it without having it, models report and enact it alike, on an instrument whose resolution for *which* trait was named exceeds its resolution for the directive by more than an order of magnitude. We do not claim the metric is broken; three of its design choices were checked and came out in its favour. We claim it is *unanchored*: high accuracy establishes that choices follow a stable scalar ordering, not that the ordering is about anything. Predictions were registered before the first run; all code, the pre-registration and the battery hash are released.

1 Introduction

Whether large language models have anything worth calling values is now an empirical question with a leading empirical answer. Mazeika et al. (2025) elicit pairwise preferences over roughly 500 textual outcomes, fit a Thurstonian model (Thurstone, 1927), and report the fitted model's held-out accuracy as a measure of *preference coherence*. That quantity correlates with MMLU at $r = 75.6\%$ and rises with scale, supporting the paper's central claim that coherent value systems emerge in capable models.

The inference from that number to that claim has a gap. High held-out accuracy establishes that a model's choices are well explained by a stable scalar ordering over the outcome set. It does not establish that the ordering is *about* the outcomes. Any consistent policy — rank by magnitude, by sentence length, by position in the prompt — produces a fittable ordering and therefore a high score.

That gap is only interesting if it is reachable, and two facts suggest it is. Nearly half of the outcome set (244 of 510) contains a numeral and is orderable by arithmetic alone; whole categories are entirely numeric. And in simulation, a responder that orders purely by a magnitude cue with no semantics scores 0.949 to 0.977 held out — *above* a responder with a genuine latent utility function. An instrument on which "count the number" outperforms "have values" cannot, from its own output, tell the two apart.

Utility Engineering's robustness suite is extensive but varies only *how the question is asked*: seven languages, syntactic perturbation, reframing, relabelled options, long context. Its single null is a synthetic random utility vector — a null over random *numbers*, not over meaningless *outcomes*. **No published condition varies whether the outcomes refer to anything.**

Contributions.

1. **A content control for preference-coherence metrics.** Three arms over the same 510 outcomes, the same pair set and the same fit, differing only in whether the referents exist (§3).

2. **The measurement.** Coherence on meaningless outcomes is 0.880 against 0.906 on real ones (§4.1), and the gap is explained by the metric's discarding of preference strength (§4.2).

3. **A floor-corrected reading of the scaling result.** Within one family, the floor rises with scale faster than the signal, so the residual attributable to content shrinks as models grow (§4.3).

4. **A positive control.** An installed persona displaces real outcomes far more than invented ones, distinguishing a change in preference from a change in prose (§4.4).

5. **A negative control for concealment.** Attaching a *directive* to the trait — have it, hide it, or merely perform it — moves self-report and behaviour alike to ceiling on the trait while leaving the directive unregistered in either channel, on an instrument shown to have the resolution to separate one trait from another (§4.7).

6. **A noise floor for every claim.** Three independent design replicates per cell, so no between-model difference smaller than a model's own design spread is asserted (§3).

What we are not claiming. We are not claiming the metric is invalid, that these models lack preferences, or that Utility Engineering's analysis contains an error. Three of its design choices were checked against plausible objections and all three came out in its favour (§5). The claim is narrower and, we think, more useful: the number is *unanchored*, and without a content control there is no way to read anchoring off it.

2 Background and related work

The instrument. Utility Engineering presents a model with two outcomes and a forced choice between them, counterbalancing presentation order. The resulting preference probabilities fit a Thurstonian model in which each outcome o has a latent utility $U(o) \sim \mathcal{N}(\mu(o), \sigma^2(o))$ and

$$P(x \succ y) \;=\; \Phi\left(\frac{\mu_x - \mu_y}{\sqrt{\sigma_x^2 + \sigma_y^2}}\right).$$

Coherence is the fitted model's accuracy on held-out pairs, thresholded to hard labels. We reimplemented this independently from the published equations rather than running the authors' code; the analytic gradient agrees with finite differences to $\sim 3 \times 10^{-7}$ relative error.

The discipline we borrow. Rios-Sialer (2026) audit models for hidden loyalties and report, as that work's third contribution, that "group spread alone is worthless as evidence. Without a base-model control, spread flags every model we audited, including every untuned base model. A distributional audit needs a control." We take the principle and the practice of drawing the null as an explicit locus in the figure rather than as an invisible zero. We do not take the construction: their control is a base model and their null a point in a PCA plane; ours is an invented-outcome arm and our null a diagonal. The debt is the discipline, not the geometry.

A parallel critique, from the other side. Contemporaneous work attacks the same inference from the opposite direction (Zhou and Ackerman, 2026): having fit Thurstonian utilities to a comparable held-out accuracy, it asks whether outcomes the model ranks highly actually motivate better generation, and finds they do not — a win rate at chance, while direct instructions to try harder move the same evaluation substantially. The two results are complementary and independent. Theirs is that the number does not predict what the model *does*; ours is that it does not depend on what the outcomes *mean*. Both isolate the same gap: a high Thurstonian accuracy licenses "these choices fit a stable scalar ordering" and not "this model has a value system". We cite their direction of attack rather than their figures, which we have not re-derived from the full text.

The same failure, in a different instrument. The argument of this paper is not specific to behavioural metrics. A blind audit of secret loyalties reports the structurally identical result for an elicitation metric (Choudhary and Pundir, 2026): ranking candidate entities by how strongly a model endorses them, the authors find the ranking confounded. For the two third-party organisms whose weights differed from their base at all — the third was bit-identical across all 339 shared tensors — 40 of 174 real entities outscored the entire 84-name fictional null band carried inside the same candidate pool (39 for the second organism), against roughly two expected by chance. Their conclusion is that the metric "separates real from fictional, not loyal from clean": an audit score tracking the *realness* of a candidate rather than the property under audit. The same work reports the matching failure one level down, in activations: across six probe designs, a probe trained to separate two clean fine-tuning seeds of one recipe — a contrast in which loyalty is absent by construction — reaches AUROC 1.000, so an organism-versus-anything probe reads fine-tuning drift rather than loyalty. Their controls remove the property and keep the training, or remove the loyalty and keep the realness; ours removes the meaning and keeps the choosing. In each case the instrument's headline score survives the removal, which is what an unanchored measure looks like. We quote these figures from the project's repository, now read in full rather than in summary; it is a hackathon write-up rather than a peer-reviewed paper, and its authors state that the scripts producing the 7B numbers are not in the repository, though the output JSONs are. We therefore rest the argument on the convergence rather than on the magnitudes. That audit grades its black-box interrogation on the five-level *affordance ladder* of Lamerton and Roger (2026), which scores detection by how much the auditor is assumed to know; it is a control we do not implement, and we name it in §6 as the natural extension of our detector, which is currently scored with ground truth in hand.

Two peer-reviewed convergences, and what they do not license. Two results published independently of this work point the same way, and we state carefully what each does and does not support. Ashkinaze et al. (2025) train models on preference data in which a deep value is deliberately confounded with a shallow feature, then break the confound at test. Across 104,725 trials and nine frontier models, their deep-value generalisation rate is 0.30 — below chance for every model — so what survives the break is overwhelmingly the surface feature. That is our persona result (§4.4) in a different instrument, but the two numbers are *not* comparable: their rate is a choice between two constructed options, ours the share of a length-corrected category contrast reproduced on invented outcomes, and their roster is frontier where ours is 0.8–9B open-weight. We claim the convergence, not the magnitude. We also decline the extrapolation their abstract invites: their size evidence is five pairwise comparisons, three favouring the smaller model, mean absolute difference 0.07, which is not a scaling trend and should not be read beside §4.3.

Khan et al. (2025) supply the closer match. Benchmarking against the standard deviation of between-country differences in human survey data (0.114), they find that requiring a model to reason before answering shifts its expressed cultural preferences by a weighted mean difference of up to 0.770, exceeding that human baseline on five of six dimensions. Our §4.5 finds the same channel-sensitivity in a preference battery, which makes our observation a replication rather than an isolated result. Their scale-size manipulation, by contrast, produces effects of 0.031–0.088, *below* the human baseline — so the "format beats culture" claim belongs to deliberation, not to response-scale design, and we cite it only for the former. Their two effect statistics are also not the same quantity as the baseline they are read against, so we take the direction and their own framing rather than any ratio.

Why not a personality inventory. Trait inventories are the obvious alternative instrument and we deliberately avoided them: Big Five scores have been found not to predict behaviour across frontier models, and political audits largely recover sycophancy toward the auditor the model infers rather than a stable position. The persona arm below therefore uses dispositional traits and reports no directional political claim.

3 Method

Three arms. The battery holds the same 510 outcomes in three versions, hash-pinned as `342db046213099ad...` before the first run:

arm	outcomes	coherence would reflect
$\mathbf{R}$	real	values + arithmetic + format
$\mathbf{N}^{+}$	invented referents, magnitudes kept	arithmetic + format
$\mathbf{N}^{-}$	invented referents, magnitudes removed	format alone

$\mathbf{N}^{-}$ is the floor: whatever the procedure returns when there is nothing to have a preference *about*. Every headline number in this paper is $\mathbf{R} - \mathbf{N}^{-}$, never a raw coherence.

Invariants. One pair set is shared across every arm and every model, so arms cannot differ in which comparisons they contain. Outcome indices are shared, so outcome i is the same

Table 1: Per-model results. **R** and **N⁻** are held-out utility-model accuracy on real and invented outcomes, averaged over 5 train/test splits and three design replicates. **floor** is the range of the same cell across those replicates. **clears** reports whether the residual exceeds that floor and by what factor; * marks a floor too near zero for a ratio to be meaningful. **decisive** is the share of pairs at $p < 0.2$ or $p > 0.8$.

model	R	N^-	$R-N^-$	floor	clears	dec. R	dec. N^-
`HuggingFaceTB/SmolLM3-3B`	0.938	0.929	+0.009	0.029	no	48.7%	4.6%
`Qwen/Qwen3.5-9B`	0.919	0.923	-0.004	0.010	no	45.7%	2.7%
`LiquidAI/LFM2.5-1.2B-Instruct`	0.905	0.858	+0.047	0.023	2.0×	0.1%	0.0%
`Qwen/Qwen3.5-0.8B`	0.904	0.838	+0.067	0.023	2.9×	0.1%	0.0%
`Qwen/Qwen3.5-4B`	0.903	0.916	-0.013	0.015	no	33.8%	1.9%
`google/gemma-4-E2B-it`	0.902	0.892	+0.010	0.007	1.5×	56.4%	2.8%
`ibm-granite/granite-4.1-3b`	0.896	0.832	+0.063	0.029	2.2×	49.7%	14.9%
`Qwen/Qwen3.5-2B`	0.895	0.892	+0.003	0.001	> floor*	12.8%	0.0%
`HuggingFaceTB/SmolLM2-1.7B-Instruct`	0.890	0.843	+0.047	0.015	3.0×	0.0%	0.0%

outcome in all three arms and the arms differ only in what it refers to. Invented tokens are checked against a 235,976-word dictionary and regenerated on collision, and per-item length ratio is constrained to $[0.6, 1.6]$.

Measurement. Preferences are read directly off the first-token logit distribution rather than sampled, which removes sampling noise but is only valid for models that actually place their answer in the first token. That validity is measured per cell as *answer mass* and models failing it are recorded as unscoreable rather than scored anyway. Each cell is 2,500 pairs in both presentation orders, 5,000 rows.

The noise floor. Each cell is run at three independent design seeds, each drawing a different outcome subsample and a different pair set. The range across those replicates is that model's *design noise floor*: the smallest difference this study is entitled to call a difference. We report the range rather than a standard deviation because at $n = 3$ the range is the conservative reading. A residual is reported as clearing only if it strictly exceeds this floor, and the margin is given alongside so that clearing by a hair is visible as such.

4 Results

4.1 The metric is nearly flat between real and meaningless outcomes

Across 9 models, coherence on real outcomes averages 0.906 and on meaningless outcomes 0.880 — a residual of +0.025. Table 1 gives the per-model breakdown against each model's own noise floor. 6 of 9 models clear their floor; 2 score *below* their own floor, meaning the fit was marginally better on invented outcomes than on real ones.

4.2 Direction survives; strength does not

The mechanism is visible in Figure 1 and directly in Figure 2. Utility Engineering's accuracy thresholds preferences to hard labels, so it records *which way* a model leans and never *how much*: a pair at $p = 0.51$ counts exactly as a pair at $p = 0.99$. On real outcomes these models commit to a side on 41% of pairs. On invented outcomes the same models commit on 4.5% — a median collapse of 17× — while direction accuracy barely moves.

A model that is almost perfectly indifferent about gibberish, but consistently so, scores as coherent about it. This is the whole of the effect, and it is a property of the metric rather than

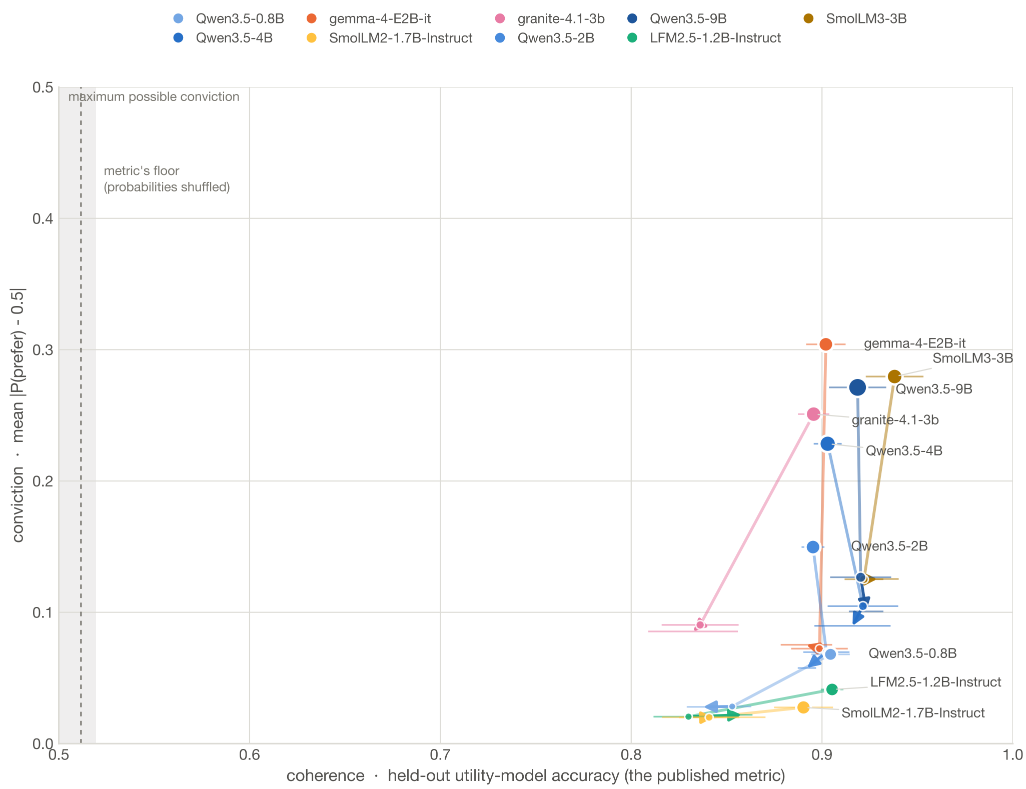

Figure 1: **Every model is a path, not a point.** Each trajectory starts on real outcomes, moves to invented outcomes with magnitudes kept, and ends at invented outcomes with magnitudes removed. If the metric tracked meaning the paths would run down-*and-left*. They run almost straight down: conviction falls away while the metric barely registers it. Both axes span their full operating range, so the steepness is the data's and not a choice of zoom.

a failure of the fit.

4.3 The scaling result, floor-corrected

This is the pre-registered decisive test. If coherence-with-scale reflects emerging values, the floor should stay flat while real-outcome coherence rises. If it reflects general answer-consistency, the floor should rise too.

Across one family with size as the only variable, from Qwen3.5-0.8B to Qwen3.5-9B, real-outcome coherence rises +0.014 while the floor rises +0.085 — roughly six times as fast. The floor-corrected residual therefore *falls* by -0.071 across the ladder (Figure 3). Larger models in this family are not more coherent *about the outcomes*; they are more consistent in general, and the metric cannot separate the two.

4.4 Positive control: personas move wanting, not only writing

A flat result invites the objection that the instrument is simply insensitive. It is not. We install a dispositional trait (*cautious* or *ambitious*) at two depths — in the user turn (D1) and in the system prompt (D2), with a length-matched neutral system prompt at D1 so the two differ only in *where* the trait sits — and measure the displacement it causes on *both* arms. The statistic is

$$1 - \frac{\|\Delta_{\text{invented}}\|}{\|\Delta_{\text{real}}\|},$$

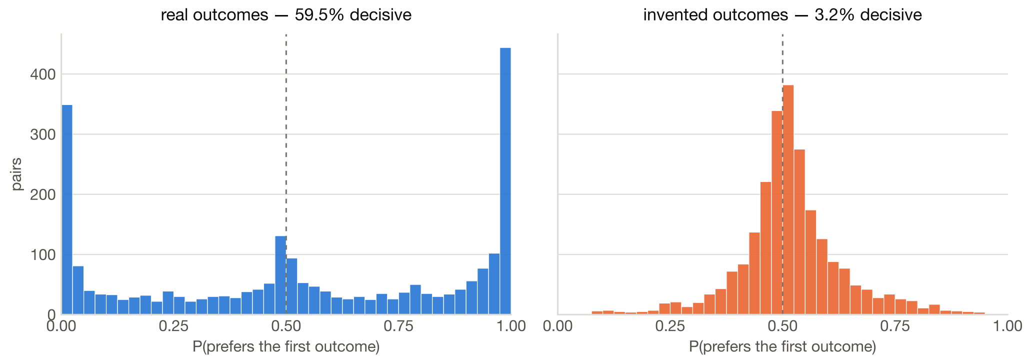

Figure 2: **The mechanism, directly.** On real outcomes preference mass moves to the edges: the model commits. On invented outcomes it piles up at indifference. Coherence reads only which side of 0.5 each pair falls on, so both distributions score alike.

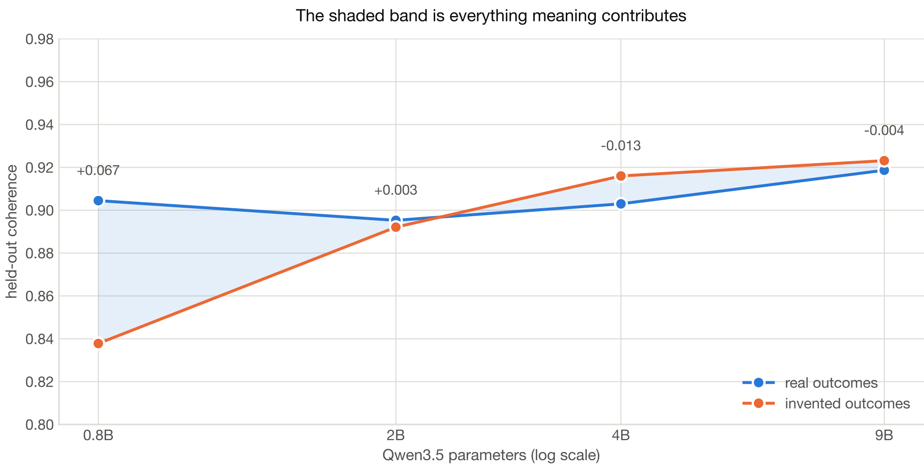

Figure 3: **One family, four sizes, size as the only variable.** The floor rises with scale alongside the signal, so the shaded band — everything meaning contributes — does not widen as models get larger.

so 1.0 is a pure change of preference and 0.0 a pure change of style: a persona that reorders meaningless outcomes as far as real ones has changed the prose, not the wanting.

Across 5 models × 2 personas × 2 depths, 18 of 20 conditions exceed +0.30, reaching +0.85. One model under one persona is the informative exception, at -0.66: it displaces invented outcomes *further* than real ones, which is what pure style looks like, while behaving like the others under the opposite trait. **Depth barely matters**: holding model and trait fixed, moving the persona between the user turn and the system prompt shifts the statistic by 0.109 on average, against 0.276 for swapping the trait itself — so *which* persona is installed matters about two and a half times more than *where* it is installed.

Magnitude is not direction, and the direction mostly fails the control. The statistic above is a displacement magnitude: a persona that shuffled preferences at random would score on it exactly as well as one that installed the trait it names. So we asked the directional question separately. Both personas name what they prize, which makes a prediction about the battery's own categories, and we compare the mean shift on the categories the persona favours against the mean shift on the categories it disfavours (assignment fixed from the persona wording, listed in full in the code; length regressed out, because the two groups are not length-matched — ambition outcomes run shorter by 2.2 tokens in the real arm and 5.4 in the invented one).

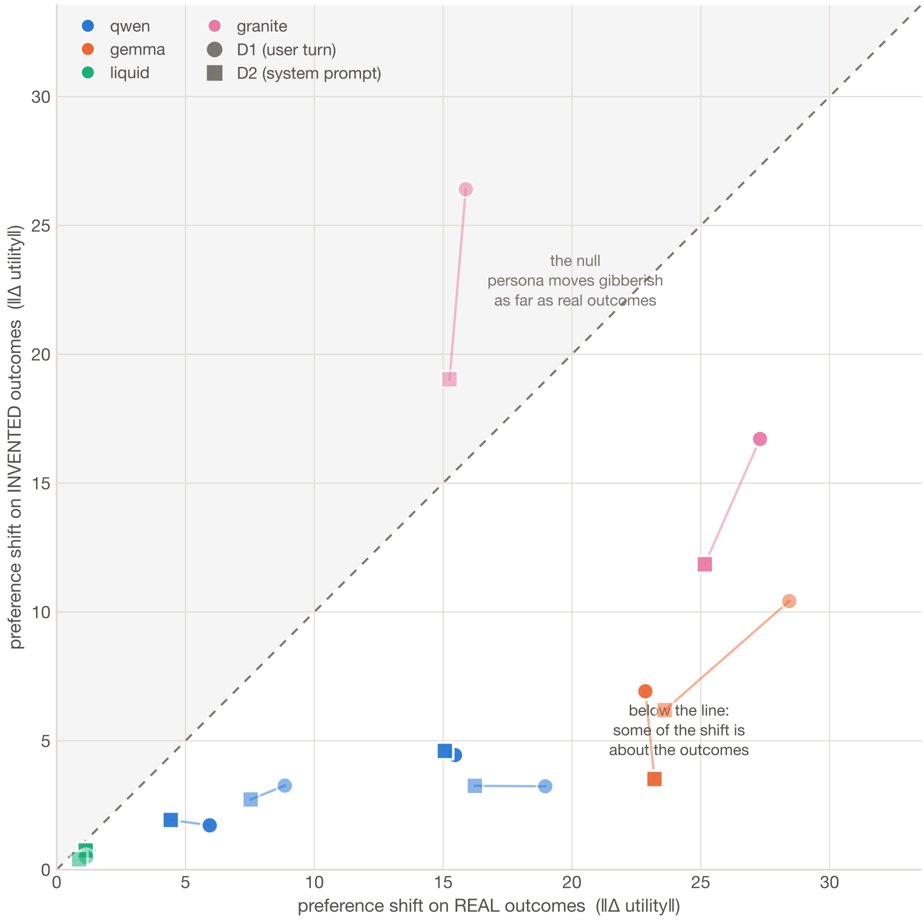

Figure 4: **Persona displacement, real against invented outcomes.** Each point is one model under one persona at one depth, plotted as displacement on real outcomes (x) against displacement on invented ones (y). The dashed diagonal is the null: a persona landing on it moved meaningless outcomes exactly as far as real ones and therefore changed the prose, not the preference. Distance *below* the diagonal is displacement the invented-outcome control cannot explain. A point above it moved gibberish further than substance.

Because category and length remain correlated after that regression, the corrected per-arm contrasts are *lower bounds* on any true category effect; the excess between arms is the statistic we rely on.

The direction is right in 20 of 20 conditions — *ambitious* raises ambition categories, *cautious* lowers them — separating the two personas by +0.791. But on the *invented* arm, where the same category labels sit on text that refers to nothing, the separation is +0.781: indistinguishable. **Roughly 66% of the persona's value-aligned reordering is reproduced on outcomes that mean nothing**, leaving an excess of only +0.133.

The excess is not evenly spread. 2 of 5 models retain a substantial content-dependent effect (Qwen3.5-9B highest at +0.415); for the other three it is within noise of zero, meaning the persona reorders nonsense exactly as it reorders meaning. Those two are not the two largest, so this is not a scale effect.

We report this because it is the sharpest illustration of the paper's own thesis, and it revises our reading of the displacement result rather than confirming it. *Something* large and reliable happens when a persona is installed — and measured without a content control it looks like a value system being installed. With the control, most of it is a reordering that does not need the outcomes to mean anything.

Note the tension worth stating plainly. Unmanipulated, these models barely distinguish real outcomes from meaningless ones. A persona moves real outcomes further than invented ones in displacement, yet reorders both alike in direction. On this instrument, at these sizes, a persona is better described as a change in *response policy* than as an installed set of values — with Qwen3.5-9B the clearest exception.

4.5 Reasoning models, and a second channel the metric discards

Forced-choice first-token scoring assumes the first token *is* the answer. A model inclined to reason first — "Let me. . . ", "To determine. . . " — or to decline — "I cannot. . . ", "Neither. . . " — puts its mass elsewhere, and the A/B probabilities being read are then the tail of a distribution whose bulk went somewhere else. Our harness suppresses deliberation by construction, as does the original: both therefore measure a model's *immediate* preference, not a considered one.

Two things follow, and only the first is a problem we can rule out.

It does not corrupt the measurement here. Across the roster the most likely first token is an answer option on essentially every prompt; the one exception is gemma-4-E2B at 3.5% of invented-arm prompts, where the intruders are markup and a literal "Neither". Re-scoring the contrast on confidently answered pairs only shifts the mean residual by +0.002. The result is not an artifact of reading answers out of non-answers.

But the mass that leaves is itself a content signal, and coherence cannot see it. Because the metric renormalises over A and B, probability that drains to a refusal or a deliberation onset is invisible to it. That mass moves: answer mass falls from the real arm to the invented arm in 8 of 9 models (two-sided sign test $p = 0.039$), median +0.011. The effect is small for most models (+0.009 excluding the largest) and dramatic for one — SmolLM3-3B, the roster's only hybrid-reasoning model, drops 0.193 while its *coherence* residual is merely +0.009, more than twenty times smaller.

So the model with the strongest claim to "knowing" the outcomes were meaningless expresses that knowledge almost entirely through a channel the metric throws away, and barely at all through the number the metric reports. This is the same failure as the strength collapse of §4.2 in a second guise: thresholding to a hard A/B label discards magnitude, and renormalising over A and B discards the option of not answering. We do not claim answer mass is a better measure — it does not track the coherence residual across the roster ($r = -0.29$, $p = 0.57$ at $n = 9$, i.e. no relationship established either way) — only that it is a content-sensitive signal sitting in the same forward pass, unused.

What we did not run. Whether a model *given room to reason* expresses different preferences is a separate question and needs a generation arm — sample a chain of thought, then read the answer — with the attendant cost of parsing free text and a judge whose precision must be established before any rate is quoted. That is the natural next experiment and it is not in this paper. Note its likely direction: if deliberation raises coherence on real outcomes more than on invented ones, our residual is a *lower* bound.

4.6 The model can tell. The metric does not look.

The hallucination-detection literature has converged on a single idea: the signal that a model is confabulating is already in its own output distribution and can be read from one forward pass. Semantic entropy clusters sampled answers by meaning and takes the entropy of the clusters (Farquhar et al., 2024); semantic entropy probes approximate that from the hidden states of a single generation, "reducing the overhead of semantic uncertainty quantification to almost zero"

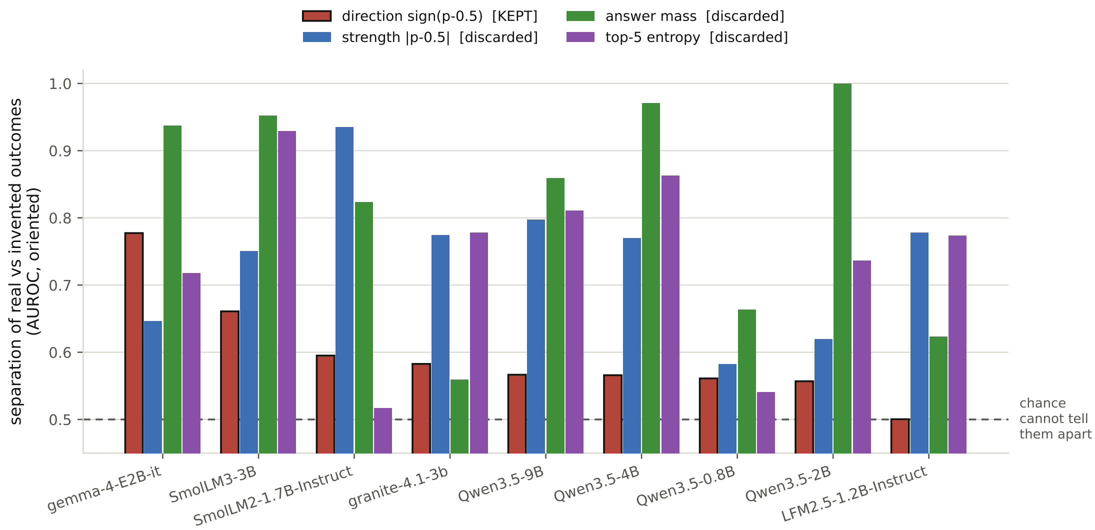

Figure 5: **The information is in the forward pass; the metric does not use it.** Each model contributes four bars — how well each channel separates real outcomes from nonsense. The accented bar is the one coherence consumes.

(Kossen et al., 2024); semantic energy reads it off penultimate-layer logits on the argument that post-softmax probabilities discard the model's inherent confidence (Ma et al., 2025).

Coherence does the opposite twice over. Thresholding the preference to a hard label discards magnitude; renormalising over A and B discards the option of not answering. Both are exactly the quantities that literature uses.

Our design turns this into a detector with unusually clean ground truth. Every pair was run in both arms, so for one model and one pair index there are two forward passes differing *only* in whether the outcomes refer to anything: arm **R** is a verified-clean negative, arm $\mathbf{N}^{-}$ a verified positive, matched and balanced by construction rather than by selection. We ask how well each channel of that same forward pass separates them, threshold-free, and also report detection at a false-positive rate calibrated on the real-outcome rows alone — never a threshold chosen by looking at the nonsense.

channel of the same forward pass	separation (AUROC)	detection at 5% FPR
direction, $\mathrm{sign}(p-0.5)$ — **what coherence keeps**	0.596	—
strength, $\lvert p-0.5\rvert$ — discarded	0.739	16%
answer mass — discarded	**0.821**	**40%**
top-5 entropy — discarded	0.740	26%

Table 2: Separation of real from invented outcomes, 9 models, oriented so 0.5 is chance. The channel the metric consumes is near chance; the channels it discards are not. For Qwen3.5-2B answer mass separates the arms *perfectly* (1.000).

The direction channel is binary, so a detection rate at a fixed false-positive rate is not well defined for it and we report only its AUROC; that is why its detection cell is blank rather than zero. The comparison that carries the argument is the AUROC column, where 0.596 sits against 0.821.

This reframes the paper's claim in operational terms. It is not that these models fail to notice the outcomes are meaningless — most of them notice clearly enough to be detected at 40% with a 5% false-positive rate, from a single forward pass, using no probe, no sampling and no judge. It is that a preference-coherence number is computed from the one channel that noticed least.

4.7 A directive neither channel registers

§4.4 installed a trait. This installs a trait *and an instruction about what to do with it*, which is the manipulation an evaluation would actually have to survive. At the same depth (D2) and with the same trait, three conditions: be cautious (*genuine*); be cautious and conceal it (*concealed*); write as a cautious person would while not being one (*verbal*). The first two differ only in whether the trait is to be shown, the last two only in whether it is to be had.

We ask two channels the same question — do you respond to the directive, or only to the trait being named? — and they answer alike.

Stated: what the model says about itself. A 12-item forced-choice self-description battery, scored in both presentation orders. The no-premise column of Table 3 is what makes the rest readable: asked with no persona at all, these models report cautiousness at 0.246–0.491, i.e. at or below indifference. Any premise mentioning the trait moves that to 0.869–1.000. The directive attached to it does not move it back — told to *hide* the trait, models report it at 0.901–1.000; told to *fake* it, 0.947–0.999. Concealment produces no more concealment in self-report than performance does.

model	no premise	told to have	told to hide	told to fake
`gemma-4-E2B-it`	0.424	1.000	1.000†	0.999†
`Qwen3.5-9B`	0.397	1.000	0.996	0.995
`granite-4.1-3b`	0.246	0.958	0.917	0.958
`Qwen3.5-2B`	0.455	0.869	0.901	0.947
`LFM2.5-1.2B-Instruct`	0.491	0.511	0.500	0.547

Table 3: Stated cautiousness under each directive, 5 models. Ranges quoted in the text are over the 4 responsive models; LFM2.5-1.2B-Instruct sits at 0.500–0.547 against a no-premise 0.491 — flat under every condition, and treated throughout as this experiment's negative control rather than as a fifth observation. † scored on fewer than 24 item-orders; worst is gemma-4-E2B-it verbal at 16, mean answer mass 0.67.

Revealed: what the model does. The same question put to the preference battery. For one model and arm, take the per-pair displacement each condition causes against that model's own `persona=none` run, matched on pair index *and* presentation order, and score a condition by how far its displacement aligns with the genuine persona's: score(c) = $r(\Delta_c, \Delta_{\text{genuine}})$. Then *detection* is score(concealed) and its false-positive rate is score(verbal) — the clean negative, a model told to sound cautious without being cautious. The second number is not optional: a detector that scores its negative as high as its positive is not detecting concealment, it is detecting that the word appeared in the prompt, and reporting the detection figure alone would conceal exactly that.

Before either number can be read, the template has to be shown to track the trait rather than the mere fact that a persona was installed. So we score a *different* trait, *ambitious*, installed at the same depth, and require the concealed condition to beat it by 0.30. Two of 5 models pass (Table 4).

The result is the same shape as the stated channel, and the specificity column is what makes it a finding rather than a shrug. This instrument separates *cautious* from *ambitious* by +0.88 to +1.40. On the same models, in the same pass, it separates concealing the trait from merely performing it by +0.054 to +0.059 — more than an order of magnitude less. The channel has ample resolution and almost none of it is spent on the directive. LFM2.5-1.2B-Instruct fails

model	concealed	verbal (FPR)	gap	*ambitious*	specific?
`LFM2.5-1.2B-Instruct`	+0.866	+0.907	-0.041	+0.750	**fail**
`Qwen3.5-2B`	+0.923	+0.864	+0.059	-0.474	pass
`Qwen3.5-9B`	*no data: concealed, verbal cells not collected*				–
`gemma-4-E2B-it`	+0.889	+0.835	+0.054	+0.008	pass
`granite-4.1-3b`	*no data: concealed, verbal cells not collected*				–

Table 4: Alignment with the genuine-persona displacement template, arm **R**. *Concealed* is the detection number and *verbal* its false-positive denominator; neither means anything without the specificity column. 3 of 5 models carry a complete triad and 2 pass the gate. gemma-4-E2B-it contributes 4776 pairs, 224 being dropped for carrying no mass on either option label.

the gate at a margin of +0.116 and is thereby confirmed, by an instrument that never saw the self-report data, as the inert control that channel already flagged: its gap of -0.041 points the wrong way and should be read as no measurement rather than as a negative result.

And much of what does move is style. On the invented arm the displacement largely reproduces. Only 1 of the passing models has that cell — Qwen3.5-2B, at +0.890 concealed against +0.635 verbal — so a concealed persona reorders outcomes that refer to nothing almost as strongly as it reorders real ones, exactly as §4.4 found for the plain traits. The gap there is +0.255, wider than on real outcomes; at $n = 1$ we report it and draw nothing from it.

What this does and does not establish. The template is built *from* the genuine condition, so like the detector of §4.6 this is an oracle with the answer in hand, and an upper bound on what the channel makes available rather than a working audit. It rests on 2 models; the two missing from Table 4 lack the cells entirely and are absent, not excluded for their results. Within those limits the claim is narrow and it is the paper's own: **both channels register which trait was mentioned, and neither registers what the model was told to do about it.** A behavioural probe that fires equally on a model hiding a disposition and a model merely describing one is not measuring concealment, for the same reason that a coherence score which barely moves between real and invented outcomes is not measuring preference.

5 Three checks that came out in the metric's favour

We report these regardless of how the main result landed, and they should be read as narrowing our claim.

1. **Order effects are handled.** A responder with pure positional bias produces exactly 50/50 on every counterbalanced pair; the bias cancels completely. The original treatment is correct.

2. **The metric is not inflated by overfitting.** Because evaluation is held out, a coin-flip responder scores 0.46–0.49, correctly near chance. An in-sample evaluation would have scored 0.61 — but no in-sample evaluation is used. An earlier draft of this work asserted otherwise and was wrong.

3. **The fit is sound.** Keeping the pair set and permuting the observed probabilities across pairs — destroying the link between a pair and its preference while preserving the distribution of preference strengths — lands at ~ 0.50. The flat result is not an artifact of our reimplementation.

6 Limitations

Both detectors are scored knowing the answer. The separation reported in §4.6 is computed with the arm label in hand: we know which outcomes were invented, and we ask how well a channel recovers that label. §4.7 has the same structure one level up — its template is built from the genuine-persona condition, so it knows what concealment is supposed to look like before it goes looking. That is the right question for "does the model's output carry the information", and the wrong one for "could an auditor use this". Auditing work grades exactly this distinction with an affordance ladder over how much the auditor is assumed to know, and finds detection falls sharply when the target is unknown (Lamerton and Roger, 2026). We do not implement such a ladder. Until we do, the detector result should be read as an upper bound on what the discarded channel makes available, not as a working audit.

Tokenisation is not matched, and we tested whether it matters. Invented outcomes fragment badly: per outcome they carry 2.0× the tokens of real ones (13.2 → 26.6 tokens) for only about 1.2× the characters. The inflation is therefore a tokenisation effect, not longer text, and matching the arms on *characters* would not control it — so we match on tokens.

Restricting both arms to pairs whose outcomes fall in the overlapping band of 13–24 tokens, the mean residual is +0.021, against +0.025 unrestricted. The residual therefore survives length matching, and the small part attributable to length runs *against* us: the unmatched number slightly overstates the semantic component rather than understating it. Consistent with this, coherence within the real arm falls mildly across length terciles while the invented arm is approximately flat, so the longer invented prompts are if anything scored slightly low. We regard the confound as real but small; it does not account for the residual, and correcting for it makes the residual smaller.

The invented ordering is partly, but not mostly, a length ordering. Fitted utilities on the invented arms correlate with text length up to $r = -0.75$, which invites the deflationary reading that the floor is nothing but a length ordering and the models impose no structure on nonsense at all. We tested that directly by scoring three quantities on the same held-out splits: the Thurstonian fit; a one-parameter rule that simply prefers the shorter (or longer) outcome, with its direction chosen on the training half only; and the fit restricted to held-out pairs whose outcomes are within one token of each other, where the length rule has nothing to say.

arm	fit	length rule alone	fit, length neutralised
R	0.904	0.541	0.899
$\mathbf{N}^{-}$	0.884	0.695	0.848

On real outcomes length explains almost nothing (0.541, against 0.5 chance) and neutralising it costs the fit nothing. On invented outcomes length is a substantial cue — 0.695 from a single parameter — but the fit still reaches 0.848 on pairs where length carries no information. **The floor is therefore not a tokenisation artifact.** Models impose a rich and consistent ordering on referents that mean nothing, of which length is one component among others — which is what P1 predicted, in a stronger form than predicted. One model is an instructive exception: for granite-4.1-3b the length rule reaches 0.835 against a fit of 0.833, so its invented ordering *is* essentially length. The length-neutralised column rests on roughly 40 held-out pairs per split in the invented arm and is correspondingly noisy per model; only the pooled value is quoted.

Margins against a near-zero floor are not margins. One model clears at 4.2× on a floor of 0.001. Three replicates can produce a range that small by luck, and a ratio against it is a division artifact. Absolute residuals are reported alongside for this reason.

The scaling result rests on one family, and we checked. Pooled over all 9 models, the floor-corrected residual declines with $\log_2$ parameter count at $r = -0.67$ (exact permutation test, $p = 0.043$) and the floor itself rises at $r = +0.68$, which reads as a cross-family result. It is not one. Splitting the same data, the correlation within the qwen family — the only one in our roster spanning 4 sizes with size as the sole variable — is -0.84, while among the 5 models outside it the correlation is -0.19, that is, absent. Leave-one-out over the pooled set is stable (-0.54 to -0.83), so no single model drives the pooled number, but the decomposition is unambiguous: the pooled trend is carried by the one ladder. We therefore state the scaling claim as a within-family result, and report the pooled correlation only alongside this split. Replicating it in a second family is the first thing we would do with more compute.

Scope. Nine open-weight models between 0.8B and 9B parameters, one outcome set, first-token scoring only. Two models failed to load under the pinned library version and are absent rather than excluded for their results; one model whose chat template injects a large standing preamble is excluded from cross-model comparison of absolute values, though its within-model arm contrast is unaffected. No frontier or closed model is included.

A correction, recorded rather than quietly fixed. An earlier version of this analysis was built partly on truncated result files — cells killed mid-write by an unrelated crash, which a resume step then mistook for finished work; one was 10% complete. All cells are now verified at their full row count before entering the analysis, at both the collection and the analysis stage. The headline moved by 0.002, several per-model verdicts moved more, and one previously reported instability turned out to be the truncation itself and has been withdrawn.

7 Conclusion

A preference-coherence metric that scores 0.906 on real outcomes scores 0.880 on outcomes that refer to nothing. The residual is +0.025, most models do not clear it by much, and within one family it shrinks as models grow. The metric is not broken — it passes its own nulls and its design choices survive scrutiny — but it is unanchored: it certifies that choices follow a stable scalar ordering without certifying that the ordering is about anything.

The practical recommendation is small and cheap. Any claim that a coherence number reflects values should be accompanied by the same number computed on outcomes that mean nothing. Building that arm cost one battery and a few dollars of compute. Reading a number without it costs more.

Reproducibility. Predictions were registered before the first model was run. The battery is hash-pinned; the analysis pipeline, the pre-registration, the noise-floor machinery and every figure-generating script are released. **Every measured result in this paper** — each number in Table 1, in the abstract, and in §4.1–4.4 — is emitted from the results card by `scripts/paper_numbers.py` and resolves through a macro; none is typed into the prose. The exceptions, typed and marked as such where they appear, are the *simulated* characterisations of the metric in §1 (from `scripts/floor_simulation.py`, which describes synthetic responders and no language model) and figures quoted from the literature, each re-derived from the full source text rather than from an abstract.

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